Integrated Forecast Diagnostics & Portfolio Analytics Framework

For Yes4All

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Agenda

Background & Objectives
Business context and project goals

Data & ETL Pipeline
Five datasets cleaned in BigQuery

Exploratory Data Analysis
Trends, SKU concentration, and performance

Cluster Analysis
Three product lifecycle segments

Forecast Diagnostics
Deviation, accuracy, bias, and alerts

Impact, Recommendations & Conclusion
$280K–$310K exposure, 5 actionable recommendations

5
Datasets

231
Products

23.7K
Transactions

3
Clusters

Background & Objective

The Challenge

Yes4All manages 200+ SKUs across fitness & wellness on Amazon, but operational data lives in silos.

Forecast deviations go undetected, leading to:

- Suboptimal inventory positioning
- Missed promotional opportunities
- Misallocated advertising spend

Project Objective

Build an integrated analytics framework delivering:

1. Daily portfolio health monitoring and SKU performance tracking
2. Automated forecast deviation alerts for proactive intervention
3. Forecast accuracy diagnostics with error and bias analysis

Problem Statement

Absence of a unified, data-driven diagnostic system that integrates sales monitoring, forecast evaluation, and product lifecycle intelligence into a single analytical framework.

1 Unstandardized Data

5 datasets with inconsistent identifiers, null formats, and data types

2 Revenue Concentration Risk

20 SKUs = 80% of GMV. Forecast failures in top SKUs propagate disproportionately

3 Uniform Lifecycle Treatment

Products follow heterogeneous trajectories but planning treats portfolio uniformly

4 Systematic Forecast Bias

SKU-specific over/under-forecasting that cannot be resolved by global model tuning

5 No Proactive Alerting

$280K–$310K annual exposure may persist undetected for weeks without alert system

Methodology

Phase 1

ETL Pipeline

5 datasets cleaned & standardized in BigQuery

→

Phase 2

Exploratory Analysis

Sales trends, SKU concentration, performance

→

Phase 3

Cluster Analysis

3 product lifecycle segments identified

→

Phase 4

Forecast Diagnostics

Deviation, accuracy, bias & automated alerts

Data & ETL (Extract-Transform-Load) Pipeline

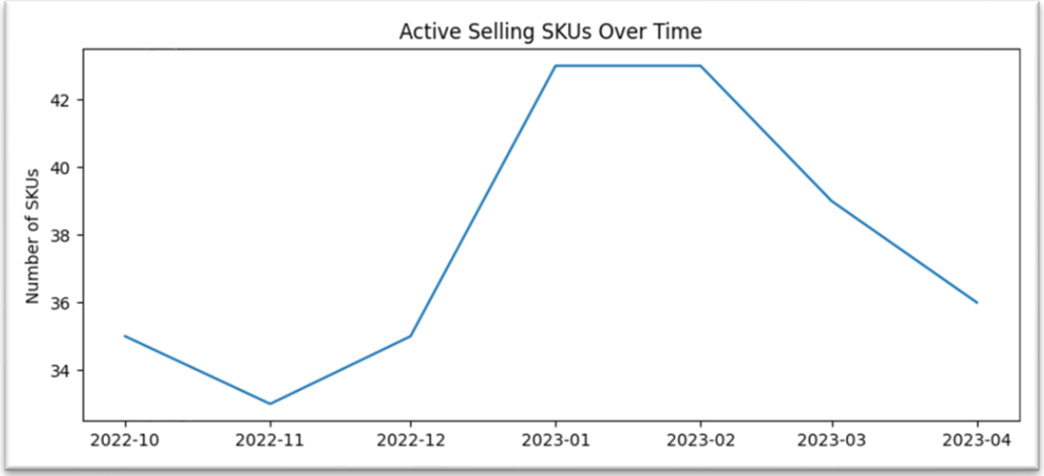
Dataset	Rows	Cols	Description
Transactions	23,716	60	Daily SKU-level sales, ads, promos
Products	231	16	SKU category & lifecycle milestones
Projection	1,812	10	Forward-looking forecast estimates
Inventory	28	33	Stock levels & incoming supply
Events	7	5	Promotional/operational events

Cleaning Transformations

- Identifiers
- Numerics
- Dates
- Validation

Exploratory Data Analysis

Active Selling SKUs Over Time

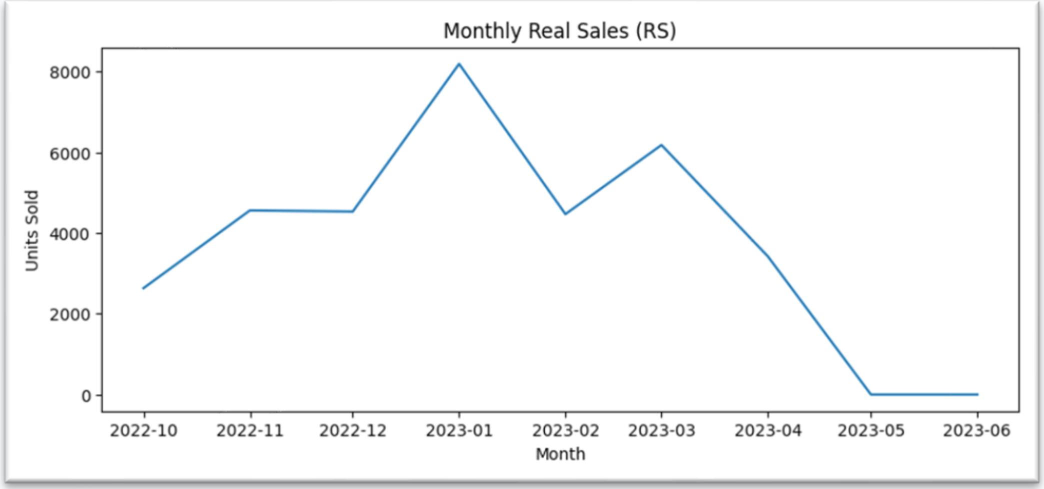


Key Insight

- The portfolio started with ~35 active SKUs in Oct 2022, dipped to ~33 in November, then expanded rapidly
- The portfolio held steady at ~43 SKUs through January and February 2023
- A clear contraction began in March 2023
- This decline directly explains the RS and GMV downturn observed in the same period
- The narrow range of the y-axis (33–43 SKUs) is worth noting

Exploratory Data Analysis

Monthly Sales Performance Trends

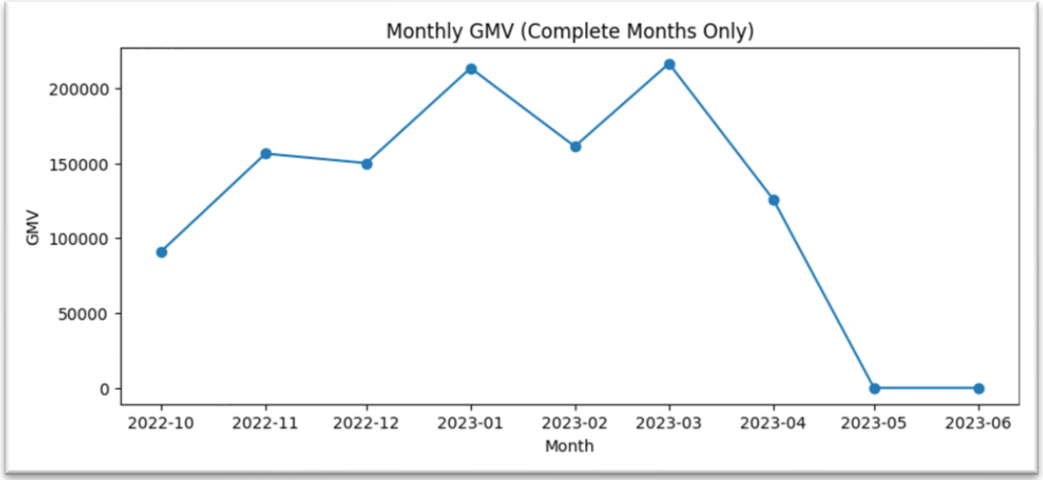


Key Insight

- Sales grew steadily from Oct 2022 through Jan 2023
- A sharp dip occurred in Feb 2023
- From March 2023 onward, sales entered a sustained downward trajectory
- May and June 2023 show near-zero sales
- Overall, the pattern suggests a portfolio that was scaling effectively through Q4 2022 into early Q1 2023

Exploratory Data Analysis

Monthly Sales Performance Trends



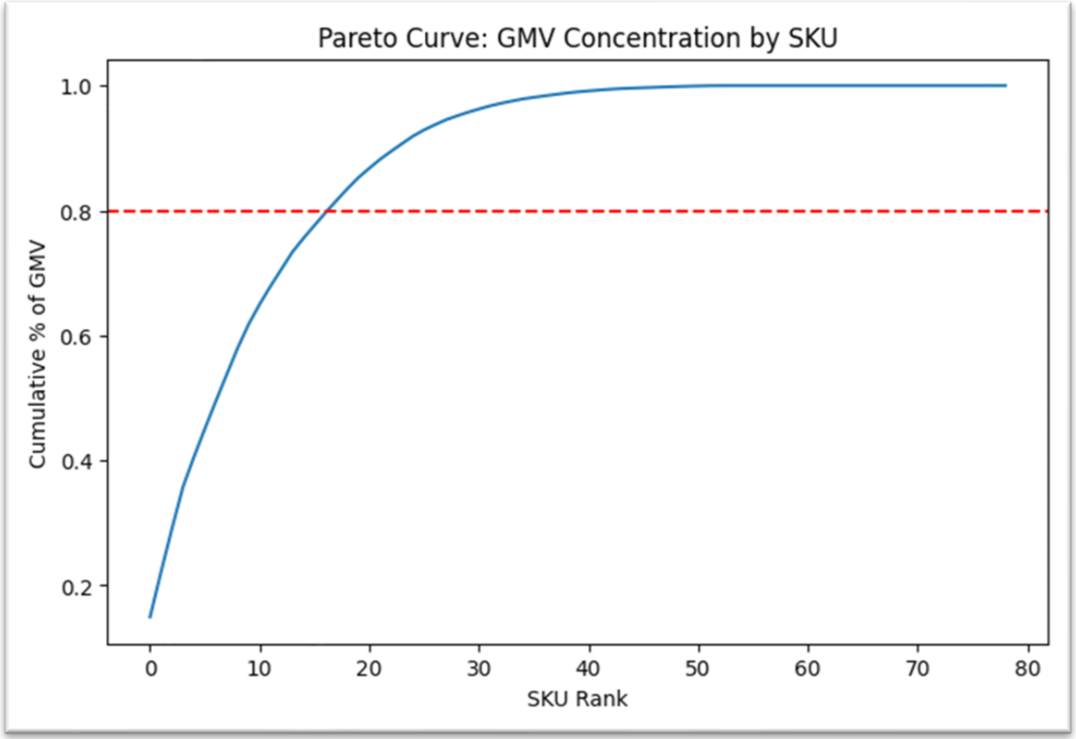
Key Insight

- GMV nearly doubled from Oct 2022 to Jan 2023
- A Feb 2023 dip to ~$162K was followed by a strong March rebound to ~$215K
- April 2023 marks the start of a steep decline
- May and June 2023 show near-zero GMV
- The GMV trend largely tracks the RS trend but with a slight difference

SKU Concentration & Performance

Pareto Analysis

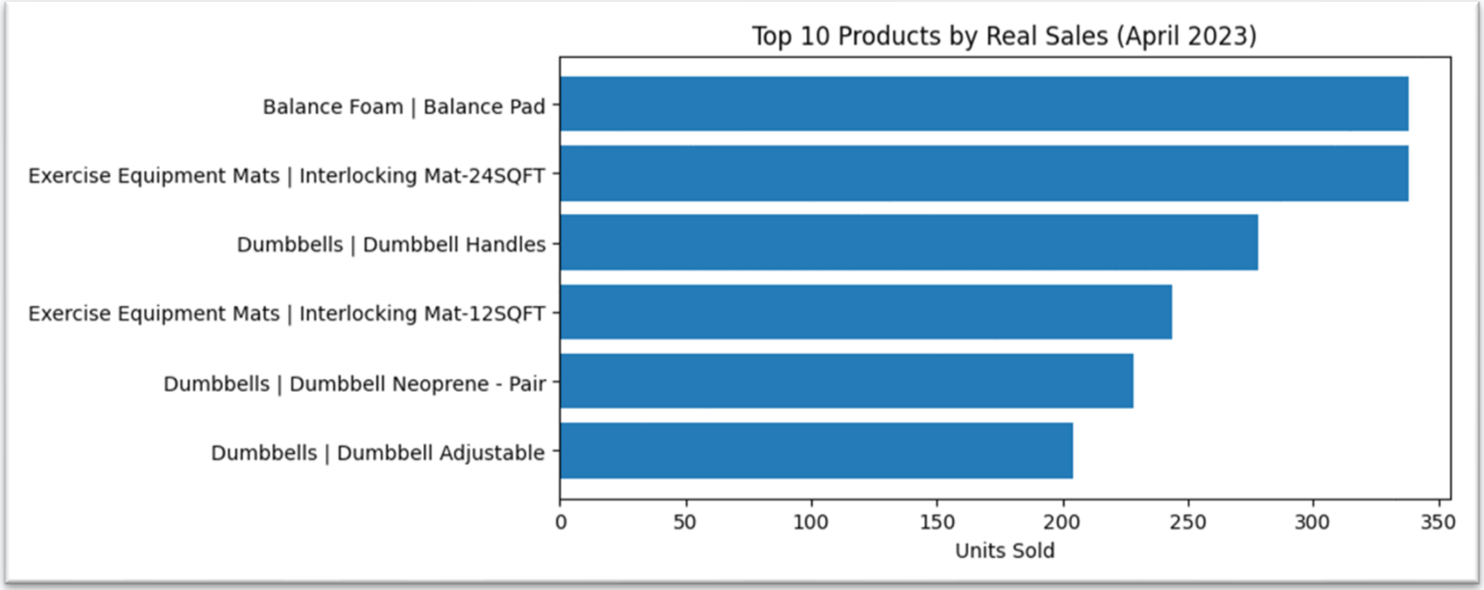
A small number of SKUs accounts for a disproportionately large share of total GMV. Revenue is heavily concentrated, reinforcing the critical importance of monitoring top-performing products.



SKU Concentration & Performance

Top Performers (April 2023)

- Balance Foam – Balance Pad
- Dumbbells — Rubber Encased
- Exercise Equipment Mats — Heavy Duty
- Dumbbells — Neoprene Pair
- Balance Foam — Balance Pad



SKU Concentration & Performance

Persistently Low Performers

- ⚠️ Dumbbell Handles- 2 units total
- ⚠️ Dumbbell Neoprene Pair- 2 units total
- ⚠️ Interlocking Mat 24SQFT — 4 units
- ⚠️ Balance Pad — 4 units
- ⚠️ Dumbbell PEVA — 4 units

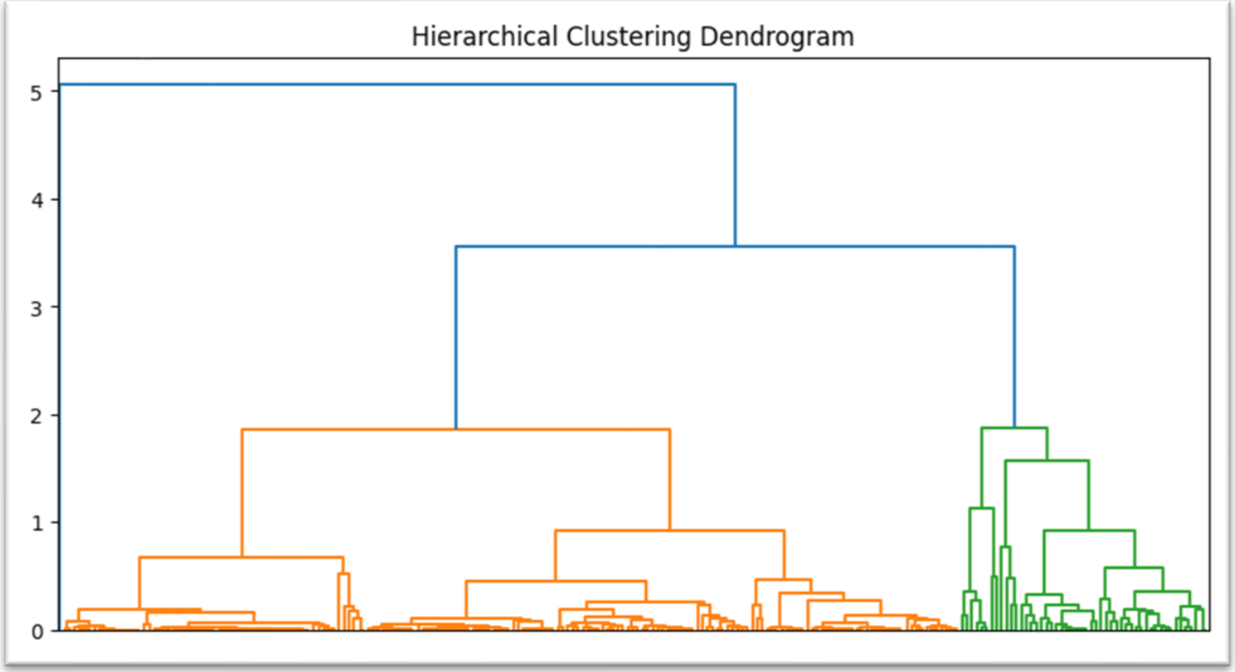
	sku	category	subcategory	rs	gmv
0	LCR2	Dumbbells	Dumbbell Handles	2	55.57
1	DOHL	Dumbbells	Dumbbell Neoprene - Pair	2	51.79
2	8F0P	Exercise Equipment Mats	Interlocking Mat-24SQFT	4	127.07
3	CAAH	Balance Foam	Balance Pad	4	118.10
4	D1MY	Dumbbells	Dumbbell Neoprene - Pair	4	175.82
5	HUJZ	Dumbbells	Dumbbell PEVA	4	426.79
6	KAL0	Exercise Equipment Mats	Interlocking Mat-12SQFT	6	144.30
7	BAAA	Dumbbells	Dumbbell Neoprene - Pair	10	698.37
8	D2CL	Dumbbells	Dumbbell Adjustable	10	732.12
9	DS24	Dumbbells	Dumbbell Neoprene - Pair	10	191.46

Cluster Analysis

Segmenting the product portfolio by lifecycle patterns

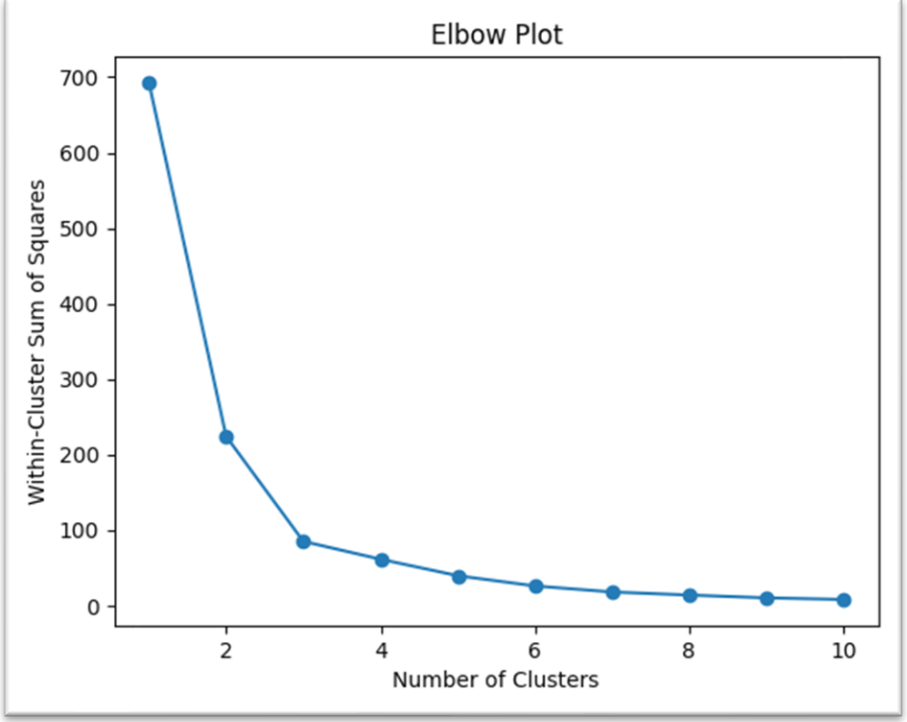
Features Selected

- Product Age (days)
- Launch → First Order (days)
- Launch → First Shipment (days)



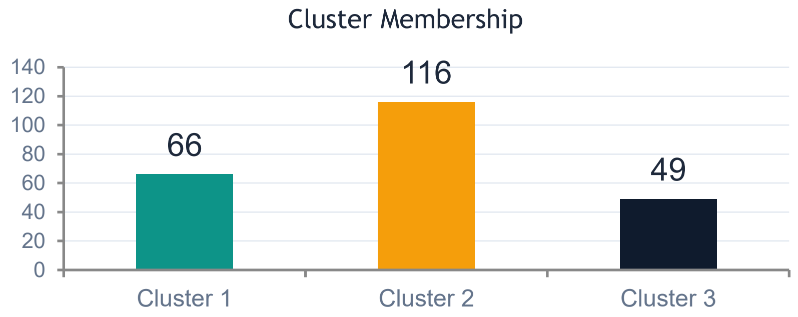
Cluster Analysis

Segmenting the product portfolio by lifecycle patterns

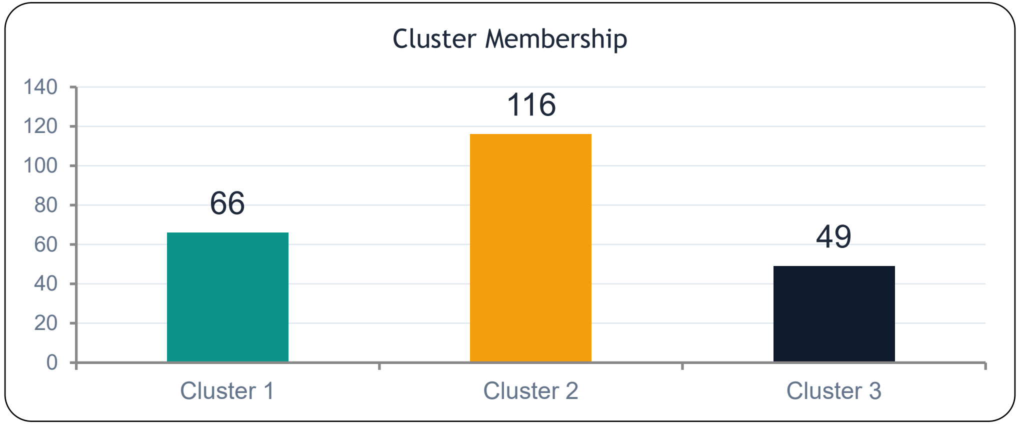


Approach

- Median imputation for missing values
- Z-score standardization
- Hierarchical clustering + Elbow method → k = 3
- K-Means with 10 random initializations



Cluster Analysis



Cluster	Product Age (Days)	Launch to First Order (Days)	Launch to First Ship (Days)
1	2454.8	156.1	156.3
2	668.9	5.2	7.0
3	2569.2	811.0	790.7

Forecast Deviation Analysis

Relative Deviation

Distribution is highly right-skewed with concentration near zero.

Most forecasts are directionally accurate, but severe deviations arise in low- or zero-demand periods where even small absolute errors create large relative errors.

Implication:

Relative deviation should be interpreted alongside absolute error and rolling baselines — not in isolation.

Absolute Deviation

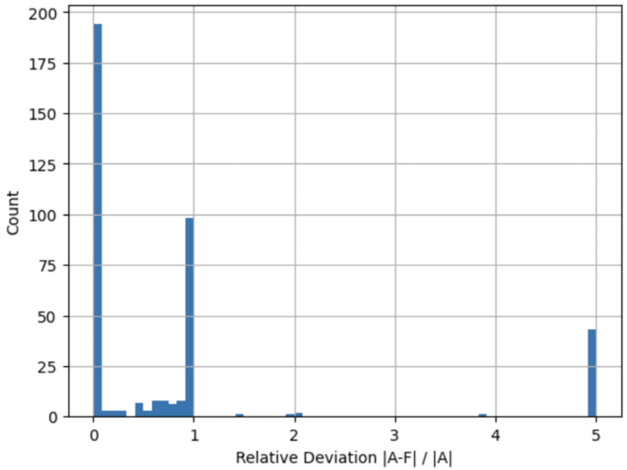
Most SKU–month observations show small absolute errors, but a heavy right tail reveals concentrated risk.

A small subset of SKU–month combinations accounts for a disproportionate share of total forecast error.

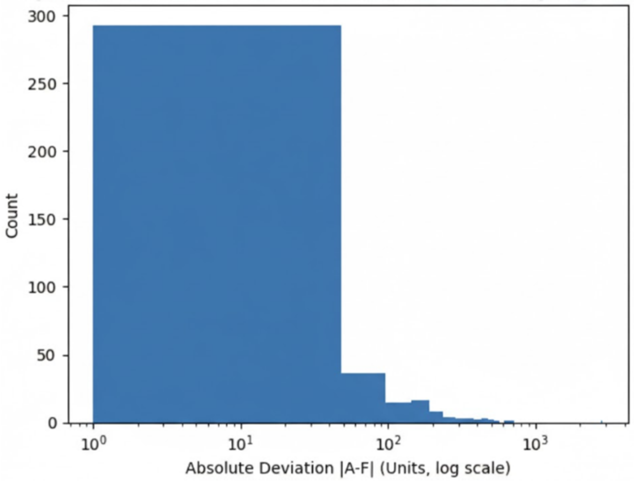
Implication:

Average accuracy metrics mask operationally significant failures. Focus diagnostic efforts on high-impact outliers.

Forecast Deviation Analysis



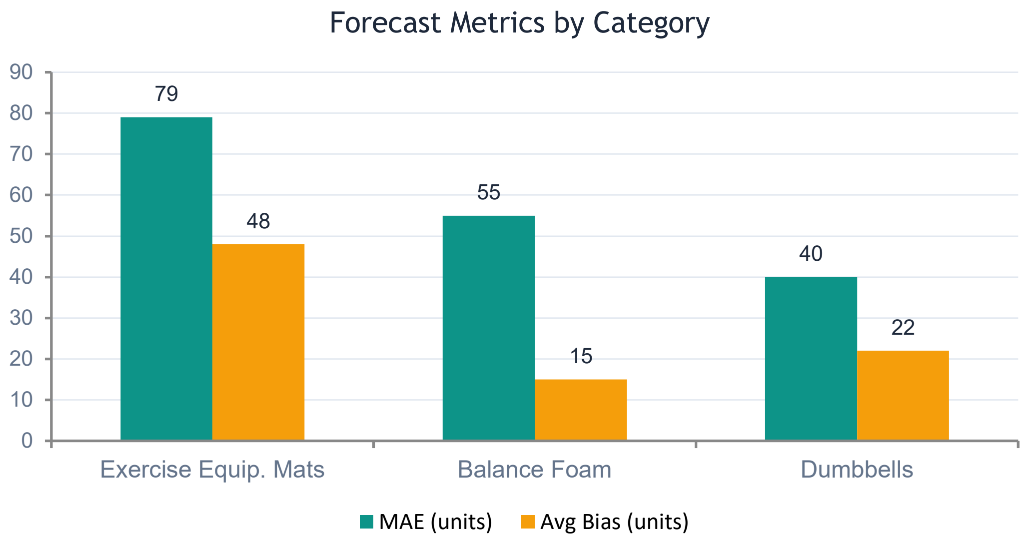
Relative Deviation



Absolute Deviation

- Relative deviation is highly right-skewed, with most observations clustered near zero
- Extreme percentage errors occur mainly in low-demand periods.
- Absolute deviation (log scale) reveals concentrated operational risk in a small subset of SKU–months
- Forecast risk is not evenly distributed — it is driven by high-impact outliers

Forecast Accuracy & Bias



Exercise Mats

Highest MAE (~79 units) with strong positive bias (+48). Consistent demand underestimation — model may miss growth/seasonality.

Dumbbells

Lowest MAE (~40 units) and small bias (+22). More stable demand with better forecast reliability.

SKU-Level Finding:

MAE ranges from 1.5 to 500+ units. Some SKUs show persistent under-forecasting (bias < −60 units, risking stockouts) while others over-forecast (bias > +450 units, increasing holding costs). Errors are systematic and SKU-specific.

Forecast Deviation by SKUs

SKU	Category	MAE (units)	RMSE (units)	MAPE (%)	Bias (units)
9W50	Dumbbells	18.3	22.7	34.2	+12.4
BAAA	Dumbbells	8.1	10.5	62.8	+5.7
BL9S	Exercise Equip. Mats	142.6	178.3	48.1	+89.3
CAAH	Balance Foam	52.4	67.8	71.5	–35.2
D2CL	Dumbbells	6.9	9.2	118.4	+4.8
HUJZ	Dumbbells	3.2	4.1	145.2	+2.1
K4VS	Exercise Equip. Mats	87.4	102.6	39.7	+61.8
KAL0	Exercise Equip. Mats	95.3	118.9	55.3	+72.4
LCR2	Dumbbells	1.8	2.4	210.6	+1.2
SLE1	Exercise Equip. Mats	68.2	84.5	42.6	+48.7
T7LS	Exercise Equip. Mats	112.8	139.4	51.4	–67.3
DS24	Dumbbells	5.4	7.1	89.3	–3.8

Forecast Deviation by Category

Category	SKU Count	Avg MAE (units)	Avg RMSE (units)	Avg MAPE (%)	Avg Bias (units)	Bias Direction
Exercise Equip. Mats	38	79.4	98.7	47.8	+48.2	Over-forecast
Balance Foam	12	55.1	68.3	64.2	–18.6	Mixed
Dumbbells	42	40.3	52.1	85.4	+22.1	Over-forecast

Automated Alert Framework

Shifting from static reporting to continuous, decision-driven monitoring

▲ Actual > Forecast

Signals unmet demand, under-forecasted growth, or emerging trends.

Action:

Expedite replenishment, review lead times, adjust forecast upward.

▼ Actual < Forecast

Signals demand deterioration, forecast overestimation, or channel disruption.

Action:

Slow replenishment, review promotional assumptions, reassess demand.

Design Features

- Rolling SKU-level baselines (dynamic, not static thresholds)
- Dual-level evaluation: SKU + Category
- Event-aware alerting — filters promotional uplift from true forecast failure

Rolling Baseline Alert Configuration

Parameter	Value	Rationale
Rolling Window	3 months (12 weeks)	Balances stability and responsiveness
Threshold Multiplier	1.5σ above rolling MAD	Targets 12–18% alert rate
Minimum Observations	4 weeks of history	Prevents alerts on newly launched SKUs
Event Exclusion Window	±7 days from event date	Filters promotional distortions
Alert Granularity	SKU × Month + Category × Month	Dual-level for tactical and strategic use

Alert Back-Testing Results

Metric	Value	Interpretation
Total SKU–months evaluated	312	Holdout period observations
Alerts triggered	47 (15.1%)	Within target range (12–18%)
True alerts (confirmed)	39 (82.9%)	Sustained trend or actionable deviation
False positives	8 (17.1%)	Transient fluctuation, reverted next month
Missed alerts (false negatives)	11	Deviations below threshold but operationally significant
Precision	82.9%	Share of triggered alerts that were actionable
Recall	78.0%	Share of actual significant deviations captured

Alert Validation & Forecast Reconciliation

Forecast-Actual Reconciliation

- 1,812 projection rows merged with Transactions
- 1,247 matched SKU–month observations (68.8%)
- 843 with non-zero actuals (67.6%)
- 404 with zero actuals despite forecasts (32.4%)
- Nov 2022–Apr 2023: highest match rates (>75%)

Back-Testing Results (Apr-Jun 2023)

82.9%	78.0%	15.1%
Precision	**Recall**	**Alert Rate**
39 of 47 alerts actionable	50 of 64 deviations caught	47 of 312 SKU-months flagged

Why 3 months? 1-month too volatile (>40% alert rate) | 6-month too slow (missed Feb 2023 dip) | Threshold: 1.5σ above rolling MAD

Key Insight

5 of 8 false positives were event-adjacent (±7-day window insufficient). Widening to ±14 days would reduce false positive rate. 11 missed alerts involved gradual declines within rolling baseline tolerance — a complementary trend-detection module could improve recall.

Validation approach: 6-month training (Oct 2022–Mar 2023) + 3-month holdout (Apr–Jun 2023)

Business Impact

$115K/yr

Excess Inventory

18,400 excess units from over-forecasting at ~$25/unit. 25% carrying cost rate.

$165K/yr

Lost Revenue Risk

8,700 units of unmet demand from under-forecasting at ~$38/unit. 50% capture rate.

$78K-$195K

Alert System Value

39 actionable alerts per year × $2K–$5K correction each. Earlier intervention on failures.

Combined Annual Exposure: $280,000 - $310,000 in avoidable costs and lost revenue

Recommendations

1. Recalibrate Exercise Mats forecast (1.3× adjustment factor for persistent +48 unit bias)
2. Deploy weekly alert dashboard — daily monitoring for top 20 revenue SKUs
3. Strategic review of Cluster 3: escalate SKUs with no order within 12 months of launch
4. Assess bottom 10 SKUs (<60 lifetime units) for listing optimization or catalog removal
5. Widen event exclusion window from ±7 to ±14 days to reduce false positives

Key Findings

Portfolio Contraction Drives Decline

Sales decline in 2023 was driven by fewer active SKUs, not widespread product underperformance. Monitoring SKU activation rates is as important as monitoring per-SKU performance.

Revenue is Highly Concentrated

A small subset of SKUs generates the majority of GMV. Forecast accuracy for these top performers has a disproportionate impact on business outcomes.

Three Lifecycle Trajectories Exist

Products follow fast-to-market, moderate, or severely delayed paths. Forecast models should account for product maturity — new launches will have less reliable predictions.

Forecast Errors are Systematic

Bias is SKU-specific and directional. Category-level patterns (e.g., persistent under-forecasting of Exercise Mats) call for category-specific calibration rather than global model tuning.

Limitations & Future Work

Limitations

- Short analysis period (~1 year) limits seasonal pattern assessment
- Clustering uses lifecycle timing only — no sales performance outcomes
- Only 68.8% of projection records matched to actuals
- Events dataset has only 7 records — may not fully capture promotional influence
- Back-testing covers 3-month holdout; longer validation would strengthen confidence

Future Directions

- Extend to multi-year data for seasonal calibration
- Integrate cluster segments into forecast models
- Build production-grade alerting dashboard
- Add predictive demand modeling capabilities

Thank You

Questions & Discussion

Patrick Nguyen